
ZipClip: Edit Less, Post More

A PROJECT REPORT (PHASE-1)

by

ABHIJITH R. (VJC22CSD003)

ANIRUDH K B. (VJC22CSD012)

JESVIN J. (VJC22CSD032)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND DESIGN

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY



DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

VISWAJYOTHI COLLEGE OF ENGINEERING AND

TECHNOLOGY, VAZHAKULAM

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Mrs. Sabitha Raju

Head of Department, CG



**DEPARTMENT OF COMPUTER SCIENCE AND DESIGN
VISWAJYOTHI COLLEGE OF ENGINEERING AND
TECHNOLOGY, VAZHAKULAM
OCTOBER 2025**

VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY, VAZHAKULAM

Department of Computer Science and Design

Vision

Moulding socially dynamite Computer Science and Design engineers capable of driving technological advancements.

Mission

1. To provide comprehensive education that combines technical knowledge, creativity and critical thinking through computer science principles and design methodologies.
2. Promote research and industry collaboration to empower our graduates for addressing real world challenges.
3. To cultivate ethical values, social responsibility and a commitment to lifelong learning among students.

Programme Educational Objectives

Our Graduates

1. Shall have strong foundation in theoretical and practical aspects of Computer Science and Design principles to develop innovative solutions.
2. Shall possess effective communication, teamwork and leadership skills to collaborate with diverse stakeholders and to practice in a corporate firm.
3. Shall have potential to become entrepreneurs, innovators and pursue research in Computer Science and Design or related fields.
4. Shall have a strong sense of social awareness and stay updated with evolving design technologies.

Program Outcomes

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
3. **Design / development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and unread in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

1. Ability to integrate theory and practice to design software applications.
2. Able to apply computer science skills and design tools to analyse, design and model composite systems.
3. Ability to design and manage projects to develop a career in a related industry.

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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN



BONAFIDE CERTIFICATE

This is to certify the project report (phase-1) titled “**ZipClip: Edit Less, Post More**” is a bonafide record of the work done by **ABHIJITH R (VJC22CSD003)**, **ANIRUDH K B (VJC22CSD012)**, **JESVIN J (VJC22CSD032)** in partial fulfillment of the award of the Degree of **Bachelor of Technology in Computer Science & Design** from APJ Abdul Kalam Technological University, Thiruvananthapuram.

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DECLARATION

We undersigned hereby declare that the project report “**ZipClip: Edit Less, Post More**”, submitted for partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology of the APJ Abdul Kalam Technological University is a bonafide work done by us under the supervision of **Mrs. Sabitha Raju**. This submission represents ideas in our own words and where ideas or words of others have been included, We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Vazhakulam

Date: 22-10-2025

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ABSTRACT

ZipClip is an innovative AI-powered application developed to revolutionize the process of short-form video creation from lengthy video content. In today's digital landscape, short-form videos dominate social media platforms such as YouTube Shorts, Instagram Reels, and TikTok, making rapid content generation a vital need for creators. However, traditional manual editing is not only time-intensive but also requires creative and technical expertise that many users may lack. ZipClip overcomes these limitations by automating the highlight generation process through advanced artificial intelligence and machine learning techniques. The system employs a multi-model AI architecture that integrates scene detection, speech and silence analysis, emotion recognition, and content scoring mechanisms to identify the most engaging portions of a video. Each module contributes to the overall understanding of the video's narrative — for instance, emotion recognition detects expressive moments, while content scoring evaluates their relevance and viewer appeal. This holistic approach ensures that the extracted clips maintain the storytelling intent of the original video while enhancing engagement value. In addition to video-based highlights, ZipClip also supports photo-based storytelling, where users can upload personal or professional photos to generate dynamic story reels enhanced with AI-curated transitions, animations, and background music. This functionality extends the application's usability beyond traditional video editors, catering to social media influencers, educators, marketing professionals, and casual users alike. The output is automatically optimized for multiple platforms, ensuring correct aspect ratios, durations, and quality settings based on the destination channel. Furthermore, ZipClip features an interactive dashboard where users can preview generated content, make fine-tuned adjustments, and export final results effortlessly. By integrating automation with creative flexibility, ZipClip drastically reduces editing time, improves consistency, and enables users to focus more on storytelling rather than technical post-production tasks. Ultimately, it transforms long-form media into concise, visually appealing narratives that align with modern digital consumption trends.

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List of Abbreviations

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CLIP	Contrastive Language-Image Pretraining
CNN	Convolutional Neural Network
DCNN	Deep Convolutional Neural Network
LSTM	Long Short-Term Memory
ML	Machine Learning
MLLM	Multimodal Large Language Model
NLP	Natural Language Processing
OCR	Optical Character Recognition

Chapter 1

INTRODUCTION

In today's fast-paced digital landscape, short-form video content has become one of the most dominant forms of media consumption. Platforms such as YouTube Shorts, Instagram Reels, and TikTok have revolutionized how users create and engage with videos, emphasizing brevity, creativity, and instant impact. As a result, creators, educators, and businesses are constantly seeking efficient methods to repurpose long-form videos into concise, captivating snippets that attract more viewers and maintain engagement. However, the process of manual video editing remains time-consuming, inconsistent, and skill-dependent, often leading to creative fatigue and reduced productivity.

ZipClip aims to solve these challenges by introducing an AI-powered automation platform that intelligently identifies and extracts the most meaningful, engaging, and visually appealing moments from lengthy video footage. By integrating artificial intelligence (AI), machine learning (ML), and natural language processing (NLP) technologies, the system performs a deep analysis of the video's visual scenes, emotions, and narrative flow. The AI evaluates content based on multiple parameters, such as expression intensity, speech dynamics, scene transitions, and viewer engagement potential, ensuring that each extracted highlight retains context and storytelling coherence.

ZipClip is developed as a web-based application designed to be intuitive and user-friendly. It leverages multi-model AI architecture that includes modules for scene detection, speech and silence analysis, emotion recognition, and content scoring. The platform also supports photo-based storytelling, allowing users to transform their personal or professional images into visually rich story reels, enhanced by AI-curated effects, transitions, and background music.

In essence, ZipClip redefines modern video editing by combining the intelligence of AI with the flexibility of user-driven customization. It enables users to “edit less and post more”, ensuring that high-quality, engaging short-form content can be generated effortlessly, thereby meeting the evolving demands of today's digital content ecosystem.

1.1 Problem Statement

Long-form videos often hide the most engaging moments, but finding them manually is slow and exhausting. Many creators lack the time, skills, or resources to transform raw footage into short, polished clips. Existing editing tools are either too complex or too expensive for everyday users. Missing the right highlights means lost audience attention and reduced impact on short-form platforms.

1.2 Objective

Automate the process of extracting highlights from long-form videos using AI-based multi-modal analysis. Provide an interactive platform for creators to preview, refine, and export clips optimized for YouTube Shorts, Instagram Reels, and TikTok. Enable easy content creation by allowing users to generate story reels from their own photos with AI-driven transitions, captions, and music. Ensure clips preserve narrative flow and emotional impact, making short-form content more engaging and platform-ready. Support platform-specific formatting so clips are instantly ready for YouTube Shorts, Instagram Reels, and TikTok. Incorporate natural language prompts (e.g., “show all funny parts”) for personalized highlight generation. Ensure scalability with a modular AI design that allows adding new agents for education, sports, or event videos. Provide a foundation for future enhancements like auto-music selection, subtitles, and multi-language support.

1.3 Scope

AI-Powered Video Summarization: Automatically analyze long-form videos to detect scenes, emotions, and key moments, producing short-form highlights. **Interactive Content Creation:** Provide an intuitive dashboard where users can preview, fine-tune, and export clips optimized for YouTube Shorts, Instagram Reels, and TikTok. **Story Reel Generation:** Allow users to upload personal photos and create AI-driven reels with captions, transitions, and music. **Scalable and Extensible Design:** Support modular AI agents, making the system adaptable for future use cases like sports highlights, educational content, and event coverage.

Chapter 2

LITERATURE SURVEY

2.1 SumBot: Summarize Videos Like a Human

The research paper titled “SumBot: Summarize Videos Like a Human” introduces an intelligent video summarization framework that aims to replicate the way humans create concise and meaningful summaries from long videos. Traditional video summarization techniques often focus on visual similarity or keyframe extraction, which can result in summaries that lack narrative flow or emotional context. In contrast, SumBot is designed to mimic the cognitive process of a human editor, selecting the most relevant and contextually rich shots that effectively convey the story or essence of the original video[1].

The system integrates several stages in its workflow. First, shot splitting is performed to divide the video into smaller, manageable segments known as shots. This step ensures that the system can analyze each part independently and identify important transitions or visual changes. Next, feature extraction is carried out using deep learning models such as Inception and Convolutional Neural Networks (CNNs). These models capture both low-level visual details (like color and texture) and high-level semantic features (such as objects, scenes, and actions). The extracted features provide a strong representation of the video’s content, enabling the system to understand its structure and meaning.

2.1.1 Methodology of SumBot: Summarize Videos Like a Human

The methodology involves shot splitting to divide videos into segments, extracting features using Inception/CNN models, and applying Markov Decision Process (MDP) and Inverse Reinforcement Learning (IRL) with Maximum Entropy (MaxEnt) reward functions and policy iteration to select optimal summaries that mimic expert human editing.

2.1.2 Model Architecture

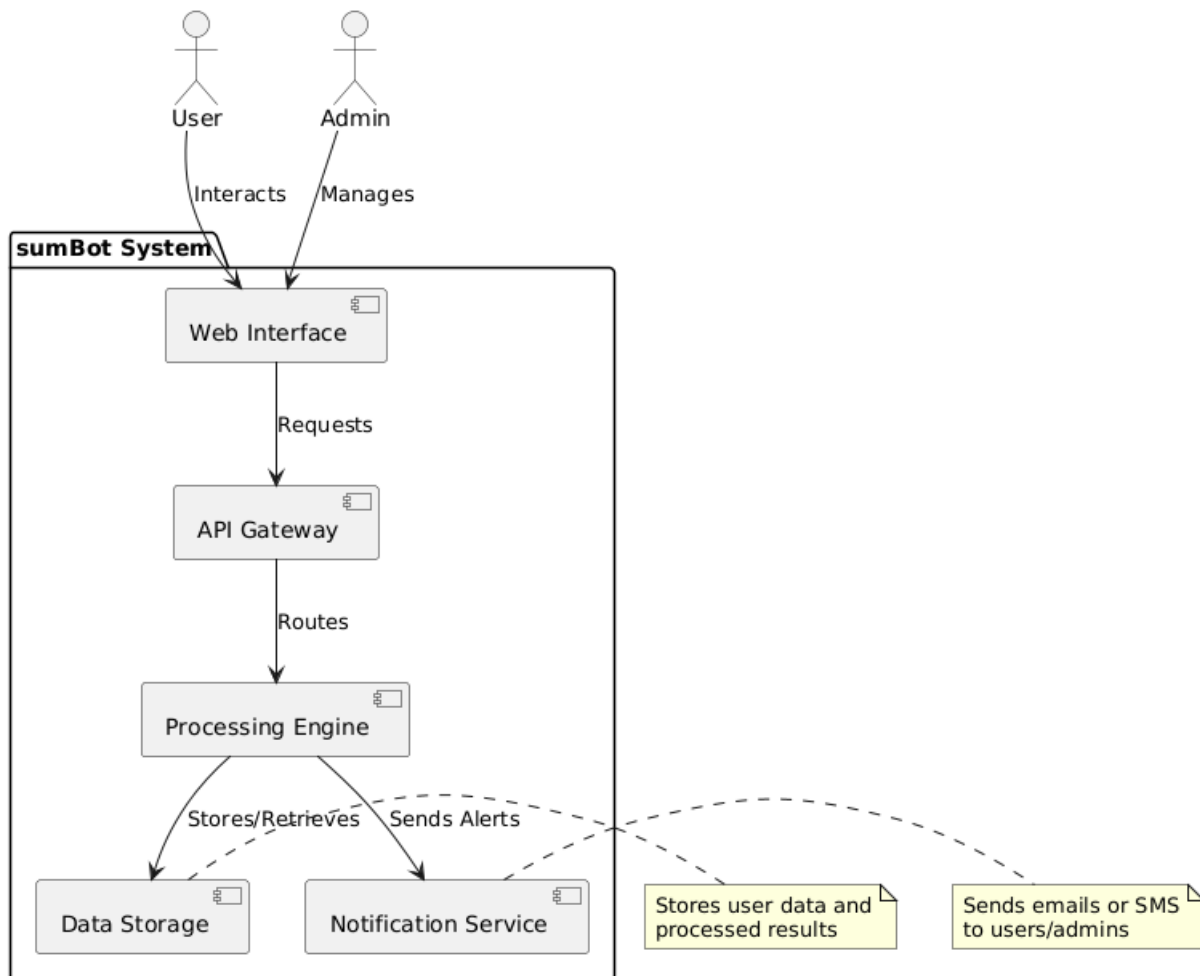


Figure 2.1: SumBot architecture

2.1.3 Advantages

1. Mimics expert human summarization with high F-score.
2. Effective feature extraction using Inception/CNN.
3. Uses MDP/IRL for optimal policy learning.

2.1.4 Disadvantages

1. Linear reward function limits complexity.
2. Focused on semi-structured videos only.
3. Soccer-focused; limited generalizability.

2.2 Video summarization using deep learning techniques: a comprehensive review

The paper titled “Video Summarization Using Deep Learning Techniques: A Comprehensive Review” provides an in-depth analysis of various deep learning approaches developed for generating concise, meaningful, and informative summaries of long video content. With the explosive growth of video data on platforms such as YouTube, Instagram, and educational archives, efficient video summarization has become a critical area of research. The review highlights how deep learning techniques have revolutionized this domain by enabling systems to automatically extract and present the most important segments of a video while maintaining temporal coherence and narrative quality[2].

The review begins by discussing the fundamental goal of video summarization—reducing the duration of a video without losing its essential information or storytelling elements. Traditional summarization methods relied on low-level visual features like color histograms and motion vectors. However, such techniques often failed to capture the semantic meaning of scenes. With the emergence of deep learning, models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs) have enabled a shift toward more context-aware and intelligent summarization systems.

2.2.1 Methodology

The methodology adopted in “Video Summarization Using Deep Learning Techniques: A Comprehensive Review” is primarily based on an extensive survey and comparative analysis of existing deep learning frameworks developed for video summarization. The authors aim to provide a detailed understanding of how different neural network architectures—especially Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models—have contributed to the evolution of this field. The review systematically explores both supervised and unsupervised approaches, evaluating their effectiveness, scalability, and suitability for various types of video data and application domains.

2.2.2 Model Architecture

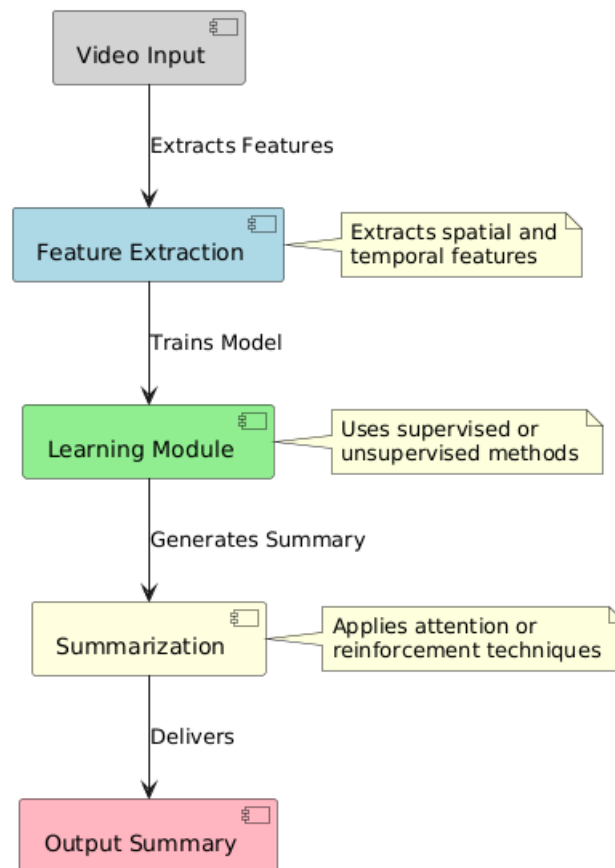


Figure 2.2: Video Summarization Review architecture

2.2.3 Advantages

1. Covers both supervised and unsupervised methods.
2. Scalable for platforms.
3. Provides a comprehensive comparison of multiple deep learning architectures like CNN, LSTM, and hybrid models.
4. Highlights recent advancements and trends, helping researchers identify future directions.

2.2.4 Disadvantages

1. Limited real-time evaluation.
2. Bias in datasets.
3. Lack of standardized benchmark datasets, making fair comparison difficult.
4. Difficulty in interpreting model decisions, reducing transparency in summarization outcomes.

2.3 Video summarization for event-centric videos

The paper titled “Video Summarization for Event-Centric Videos” presents a specialized approach to summarizing videos that revolve around specific events, such as sports matches, concerts, conferences, or social gatherings. Unlike general-purpose summarization techniques that treat all video content uniformly, this method focuses on identifying and extracting event-driven highlights that capture the key moments, actions, and emotions associated with a particular event. The goal is to generate concise summaries that not only shorten the viewing time but also retain the contextual and emotional essence of the original video[3].

The system integrates a multimodal feature fusion strategy, combining visual, audio, and sometimes textual information (such as subtitles or metadata) to achieve a richer understanding of the video content. Visual features, such as player movements or crowd reactions, are extracted using Convolutional Neural Networks (CNNs), while audio features like cheering, applause, or commentary are analyzed to detect moments of excitement or importance. This multimodal fusion ensures that the summarization process goes beyond mere visual cues and considers the broader sensory and contextual information present in the video.

2.3.1 Methodology

The methodology proposed in “Video Summarization for Event-Centric Videos” centers around an event-driven framework that emphasizes the fusion of multimodal features to extract meaningful highlights while preserving the contextual flow of the video. Unlike traditional summarization methods that rely solely on visual information, this approach integrates multiple data sources—such as visual, audio, and textual cues—to achieve a deeper understanding of the events occurring within the video.

The process begins with event detection and segmentation, where the video is divided into smaller temporal segments or shots based on significant changes in visual or audio patterns. Each segment is analyzed to determine its relevance to the main event. Convolutional Neural Networks (CNNs) are used to extract spatial and semantic features from the visual stream, capturing key elements like people, objects, or activities. At the same time, audio feature extraction techniques identify crucial auditory signals such as applause, cheering, or tone changes in speech, which often indicate moments of importance or excitement.

2.3.2 Model Architecture

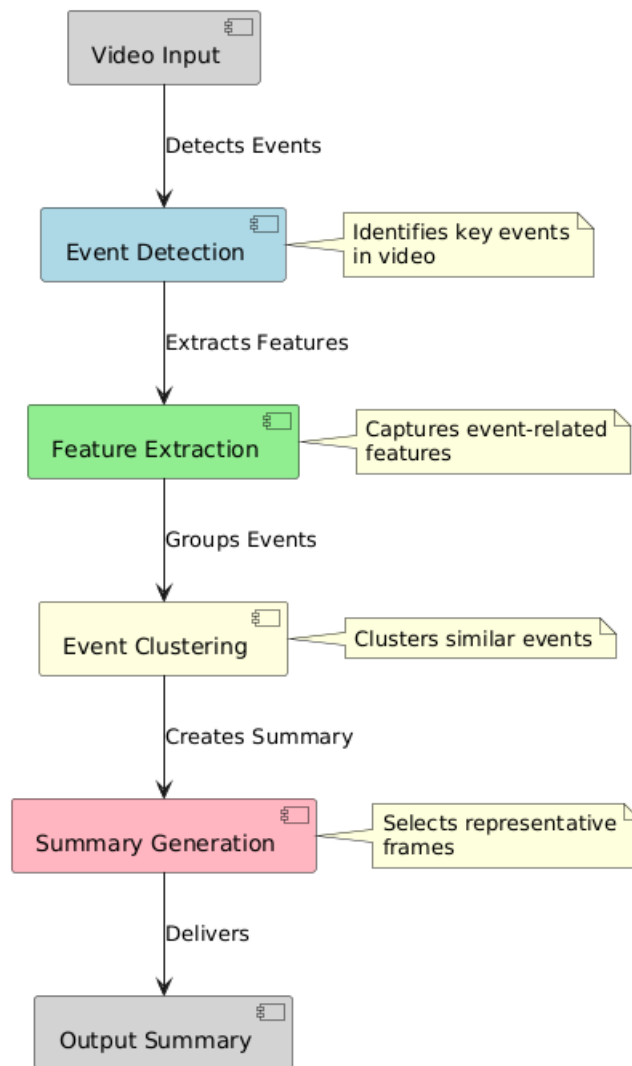


Figure 2.3: Event-Centric Video Summarization architecture

2.3.3 Advantages

1. Effective for event highlights.
2. Multimodal fusion enhances context.

2.3.4 Disadvantages

1. Not tested on very short-form content.
2. Limited domain adaptation.

2.4 AI-driven video summarization for optimizing content search and retrieval efficiency

The paper titled “AI-Driven Video Summarization for Optimizing Content Search and Retrieval Efficiency” explores the use of artificial intelligence techniques to enhance how videos are summarized, indexed, and retrieved across large-scale content platforms. With the massive increase in online video data, efficient search and retrieval have become crucial for both end-users and content creators. This research focuses on developing intelligent summarization systems that can automatically extract spatial and temporal highlights, thereby improving accessibility, searchability, and automation in content management workflows[4].

The primary objective of the system is to create summaries that not only condense video length but also serve as metadata-rich representations suitable for search engines and recommendation systems. By analyzing both spatial features (objects, faces, and scenes) and temporal dynamics (events, transitions, and motion patterns), the proposed AI-driven framework ensures that the summaries maintain semantic coherence and relevance. This allows users to quickly preview, locate, and retrieve specific moments within lengthy video archives.

2.4.1 Methodology

The methodology presented in “AI-Driven Video Summarization for Optimizing Content Search and Retrieval Efficiency” focuses on combining spatial and temporal analysis to produce intelligent video summaries that enhance both content accessibility and retrieval performance. This approach leverages deep learning models to extract, interpret, and organize meaningful video features, enabling faster and more accurate search operations across large video repositories.

The process begins with spatial feature extraction, where each video frame is analyzed using Convolutional Neural Networks (CNNs). These networks identify and encode visual elements such as objects, faces, actions, and background scenes. By capturing these spatial cues, the system generates high-level semantic representations that describe the visual content of each segment. This step ensures that key frames are not only visually distinct but also semantically informative.

2.4.2 Model Architecture

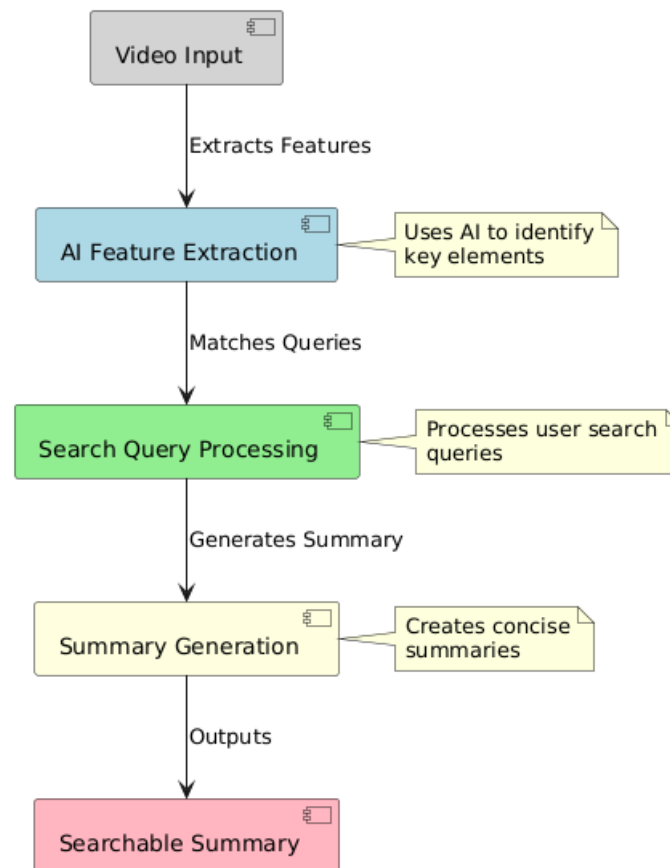


Figure 2.4: AI-Driven Video Summarization for Search architecture

2.4.3 Advantages

1. Optimized for retrieval.
2. Automation for creators.
3. Combines spatial and temporal analysis for more context-aware summarization.
4. Enhances search accuracy by indexing videos with semantic tags and feature embeddings.

2.4.4 Disadvantages

1. Focus on search, not platform deployment.
2. Evaluation on YouTube only.
3. Limited real-time performance when handling long or high-resolution videos.
4. Lacks user interaction or feedback mechanisms to refine summary quality.

2.5 Video summarization via knowledge-aware multimodal networks

The paper titled “Video Summarization via Knowledge-Aware Multimodal Networks” proposes a novel approach to video summarization that integrates domain knowledge with multimodal feature fusion to produce more robust and contextually relevant highlights. Unlike conventional methods that rely solely on visual or temporal patterns, this framework incorporates prior knowledge about the content, events, or domain to guide the summarization process, ensuring that the selected highlights are meaningful and aligned with human understanding[5].

A key innovation of this work is the use of knowledge-driven multimodal fusion. The system combines inputs from multiple modalities, including visual cues (frames and objects), audio signals (speech, music, or environmental sounds), and textual metadata (subtitles or captions). Additionally, it leverages knowledge graphs or domain-specific ontologies to provide context about relationships between objects, events, or actions. This allows the model to prioritize segments that are semantically significant rather than merely visually salient.

2.5.1 Methodology

The methodology of “Video Summarization via Knowledge-Aware Multimodal Networks” revolves around leveraging knowledge-driven deep learning frameworks to fuse multimodal video data and extract meaningful highlights. The approach integrates information from multiple sources, including visual frames, audio signals, and textual metadata, while incorporating domain knowledge to guide the summarization process and improve semantic relevance.

The process begins with multimodal feature extraction, where each modality is processed by a dedicated agent. Visual features are captured using Convolutional Neural Networks (CNNs) to identify objects, scenes, and key visual events. Audio signals, such as speech, music, or environmental sounds, are analyzed using spectrogram-based neural networks to detect moments of interest. Textual information, including subtitles or captions, is processed using natural language processing techniques to extract semantic context and relevant keywords.

2.5.2 Model Architecture

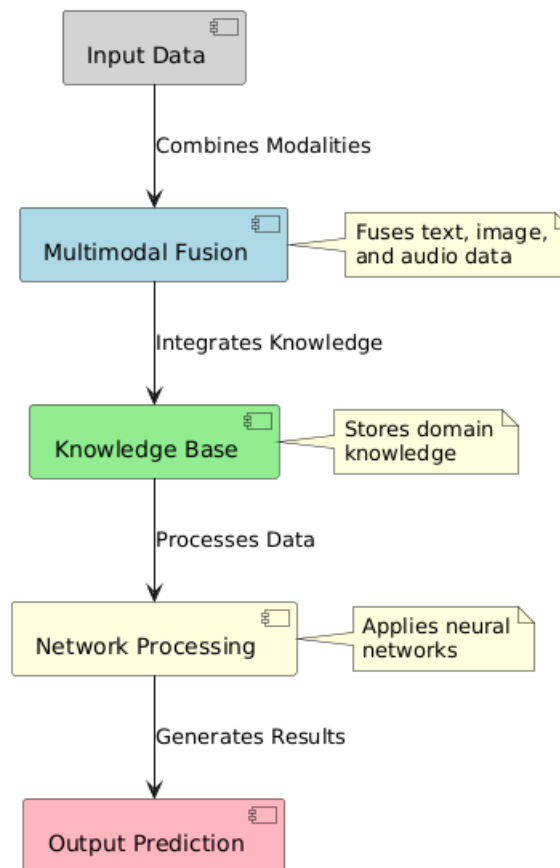


Figure 2.5: Knowledge-Aware Multimodal Networks architecture

2.5.3 Advantages

1. Robust highlight extraction.
2. Knowledge-driven fusion.
3. Maintains temporal and semantic coherence in the generated summaries.
4. Multi-agent collaboration enables better integration of visual, audio, and textual features.

2.5.4 Disadvantages

1. Computational cost.
2. Sparse real-user studies.
3. Dependency on accurate knowledge graphs or domain-specific ontologies, which may not always be available.
4. Potential delays in real-time summarization for long or high-resolution videos.

2.6 Zero-shot scene change detection

The paper titled “Zero-Shot Scene Change Detection” introduces a novel approach for detecting scene transitions in videos without requiring any prior training or labeled datasets. Traditional scene change detection methods often rely on supervised learning models, which need large annotated datasets and extensive training to recognize shot boundaries accurately. In contrast, this study presents a zero-shot framework, enabling the system to identify scene changes directly from the video content itself, making it highly adaptable to new or unseen datasets and reducing reliance on manual annotations[6].

The proposed method leverages feature differencing techniques to detect boundaries between consecutive frames or shots. Visual features are extracted from frames using pre-trained deep learning models, such as CNNs, which encode spatial information including objects, textures, and scene layouts. By computing differences between consecutive frame features, the system identifies abrupt or gradual changes that signify scene transitions. This allows for effective detection of both hard cuts and soft transitions, such as fades or dissolves, which are common in edited videos.

2.6.1 Methodology

The methodology of “Zero-Shot Scene Change Detection” focuses on identifying scene boundaries in videos without requiring any prior training or labeled datasets, making it highly adaptable to a wide variety of video content. The approach relies primarily on feature differencing, a technique that quantifies changes between consecutive frames or shots to detect significant transitions.

The process begins with frame-level feature extraction, where each frame is encoded using pre-trained deep learning models such as Convolutional Neural Networks (CNNs). These models capture rich visual information, including objects, textures, colors, and spatial layouts, providing a high-level representation of each frame. By comparing the features of consecutive frames, the system calculates the magnitude of change, which serves as an indicator of potential scene boundaries.

To improve robustness, the methodology often incorporates temporal smoothing techniques, which help to filter out minor fluctuations caused by camera movements, lighting changes, or background variations. This ensures that only significant changes—indicative of actual scene transitions—are detected. Additionally, adaptive thresholding is applied to differentiate between hard cuts, soft transitions, and gradual scene changes, allowing the system to handle a variety of editing styles commonly found in movies, TV shows, and online videos.

2.6.2 Model Architecture

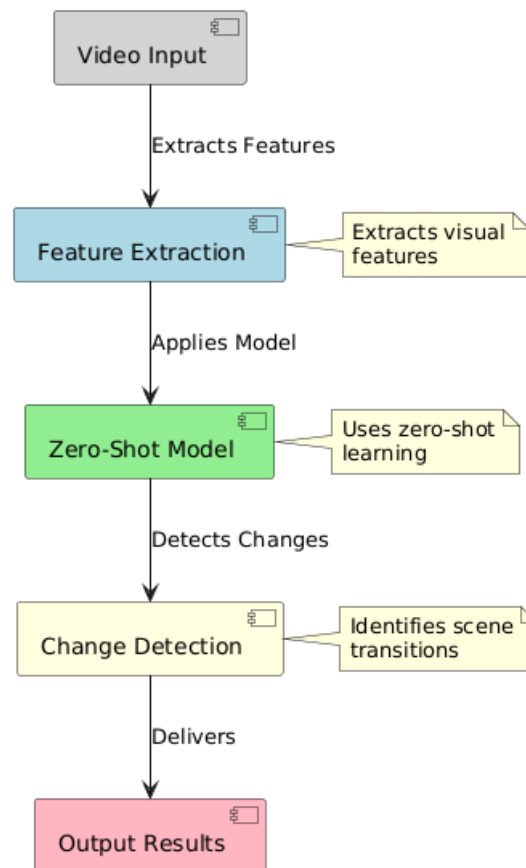


Figure 2.6: Zero-Shot Scene Change Detection architecture

2.6.3 Advantages

1. No training required.
2. Effective for boundary detection.
3. Scalable for large datasets without the need for labeled data.
4. Robust to different video genres and editing styles.

2.6.4 Disadvantages

1. May miss subtle transitions.
2. Limited evaluation on user videos.
3. Potential sensitivity to minor camera motion or lighting changes if temporal smoothing is insufficient.
4. Lacks integration of audio or textual cues, which could improve boundary detection in complex scenes.

2.7 Scene Object Change Detection and Fine-Grained Target Recognition

The paper titled “Scene Object Change Detection and Fine-Grained Target Recognition” presents a sophisticated approach to detecting changes in video scenes and identifying objects at a pixel-level granularity. Unlike traditional methods that often focus on coarse frame-level analysis, this work emphasizes high-precision segmentation and fine-grained recognition, making it suitable for applications where accurate detection of small or subtle changes is critical, such as surveillance, autonomous driving, or environmental monitoring[7].

The system employs advanced image processing and deep learning techniques to identify objects and scene changes with exceptional detail. Pixel-level change detection allows the model to differentiate even minor alterations in the scene, such as a moving object, lighting change, or small spatial adjustments. Once these changes are detected, fine-grained target recognition techniques are applied to categorize and segment each object accurately, enabling precise localization and identification.

2.7.1 Methodology

The methodology of “Scene Object Change Detection and Fine-Grained Target Recognition” revolves around achieving high-precision detection of scene and object changes at the pixel level while accurately recognizing and segmenting targets within the video. The system is designed to capture subtle alterations and provide detailed semantic information, making it suitable for applications requiring granular analysis.

The process begins with pixel-level feature extraction, where each frame is analyzed to identify low-level visual attributes such as color, texture, and edges, as well as high-level semantic features like object identity, shape, and context. Advanced Convolutional Neural Networks (CNNs) or other deep learning architectures are used to encode these features, producing rich representations that highlight significant objects and scene elements.

Once features are extracted, the system performs change detection by comparing consecutive frames or reference scenes. This allows it to pinpoint alterations in the environment, such as object movement, appearance, disappearance, or other modifications. By operating at the pixel level, the methodology ensures that even subtle changes are captured, avoiding the loss of important information that occurs with coarse frame-level analysis.

2.7.2 Model Architecture

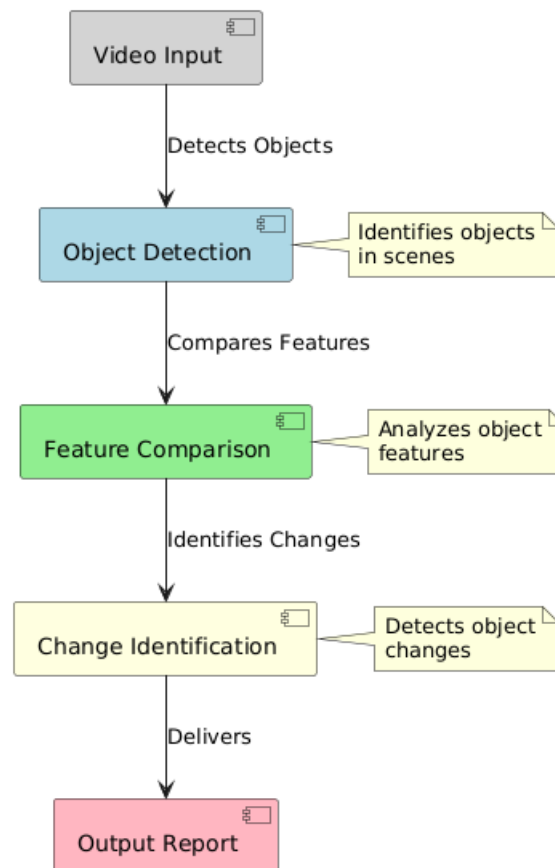


Figure 2.7: Scene Object Change Detection architecture

2.7.3 Advantages

1. High precision segmentation.
2. Pixel-level accuracy.
3. Robust for applications requiring detailed scene understanding, such as surveillance or autonomous navigation.
4. Enhances downstream tasks like video summarization, content indexing, and automated monitoring.

2.7.4 Disadvantages

1. Computationally expensive.
2. Not robust to occlusion.
3. Limited real-world evaluation, reducing insight into performance across diverse environments.
4. Potential sensitivity to visual noise, lighting changes, or occlusions.

2.8 Advances in Video Emotion Recognition: Challenges and Progress

The paper titled “Advances in Video Emotion Recognition: Challenges and Progress” presents a comprehensive review of the latest developments in video-based emotion recognition, focusing on both methodological advancements and practical challenges. Emotion recognition from video is a complex task that involves analyzing multimodal cues such as facial expressions, vocal intonations, body gestures, and contextual information to infer the emotional state of subjects accurately[8].

A key aspect highlighted in this review is the use of multimodal fusion techniques. By integrating visual, audio, and sometimes textual information, modern emotion recognition systems achieve a more holistic understanding of human emotions. Visual cues, such as facial micro-expressions or body movements, are processed using Convolutional Neural Networks (CNNs) or advanced facial landmark detection models. Audio cues, including tone, pitch, and speech rhythm, are analyzed using Recurrent Neural Networks (RNNs), LSTMs, or Transformer-based architectures. The fusion of these modalities enhances accuracy and robustness, allowing the system to compensate when one modality is ambiguous or noisy.

2.8.1 Methodology

The methodology of “Advances in Video Emotion Recognition: Challenges and Progress” focuses on surveying state-of-the-art techniques for capturing and analyzing emotions from videos using temporal modeling and multimodal fusion. Since emotions are inherently dynamic and expressed through multiple modalities, the survey emphasizes methods that combine visual, audio, and sometimes textual information to achieve robust recognition.

The process begins with visual feature extraction, where facial expressions, micro-expressions, and body gestures are analyzed using Convolutional Neural Networks (CNNs) or facial landmark detection models. These features capture subtle spatial cues that indicate emotional states, such as eye movements, lip curvature, or body posture changes.

Next, audio features such as pitch, tone, intensity, and speech rhythm are extracted using Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, or Transformer-based models. These models track the temporal evolution of vocal cues, which is critical because emotional expressions in speech often change gradually over time.

2.8.2 Model Architecture

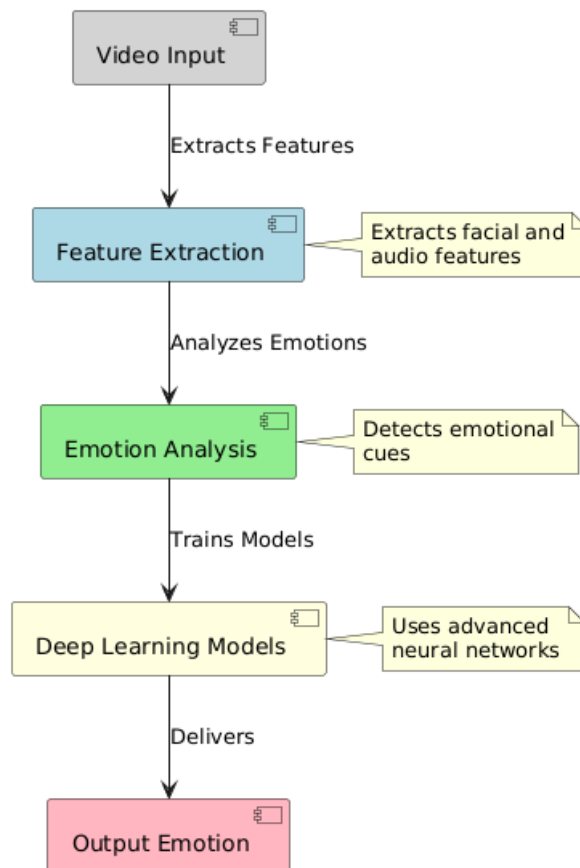


Figure 2.8: Video Emotion Recognition Advances architecture

2.8.3 Advantages

1. Multimodal fusion for better accuracy.
2. Temporal modeling enhances robustness.
3. Attention-based fusion helps focus on the most informative segments.
4. Provides insights into real-world challenges such as lighting, occlusion, and background noise.

2.8.4 Disadvantages

1. Most tests on academic datasets.
2. Limited cross-platform evaluation.
3. High computational requirements for processing multimodal and temporal features.
4. Lack of real-time performance evaluation for practical deployment.

2.9 Multimodal Emotion Recognition with Deep Learning

The paper titled “Multimodal Emotion Recognition with Deep Learning” explores the integration of visual, auditory, and textual data for accurate emotion recognition in videos. Traditional approaches often rely on single modalities, which can be insufficient when cues from one modality are ambiguous or noisy. This work leverages deep learning-based fusion techniques to combine multiple modalities, providing a more holistic understanding of emotional states[9].

A central focus of the paper is the use of cross-modal signal integration, where information from video frames, audio signals, and text (such as subtitles or transcripts) is analyzed collectively. Visual cues, such as facial expressions, micro-expressions, and gestures, are captured using Convolutional Neural Networks (CNNs). Audio features, including pitch, intensity, and tone, are extracted with Recurrent Neural Networks (RNNs) or LSTM-based models. Textual cues are processed using Transformer-based natural language processing techniques to infer semantic sentiment or emotional content.

2.9.1 Methodology

The methodology of “Multimodal Emotion Recognition with Deep Learning” is centered on integrating video, audio, and textual data using deep learning to achieve accurate and robust emotion recognition. The system is built on a modular design, where each modality is processed independently to capture the most relevant features before fusion, ensuring flexibility and scalability.

Visual features are extracted from video frames using Convolutional Neural Networks (CNNs), focusing on facial expressions, micro-expressions, gestures, and body posture. These features capture spatial cues indicative of emotional states. Audio features are obtained using Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks, analyzing tone, pitch, intensity, and speech dynamics to detect vocal emotion cues. Textual features, such as transcripts or subtitles, are processed using Transformer-based NLP models to extract semantic and sentiment information, adding another layer of emotional understanding.

Once modality-specific features are obtained, fusion techniques are applied to integrate these diverse data streams. Fusion can be early, combining features before temporal modeling, or late, combining predictions from individual modalities. Advanced approaches incorporate attention mechanisms or cross-modal transformers to model interactions between modalities, highlighting the most informative signals while minimizing redundancy. This ensures the final emotion prediction reflects a holistic understanding of all available cues.

2.9.2 Model Architecture

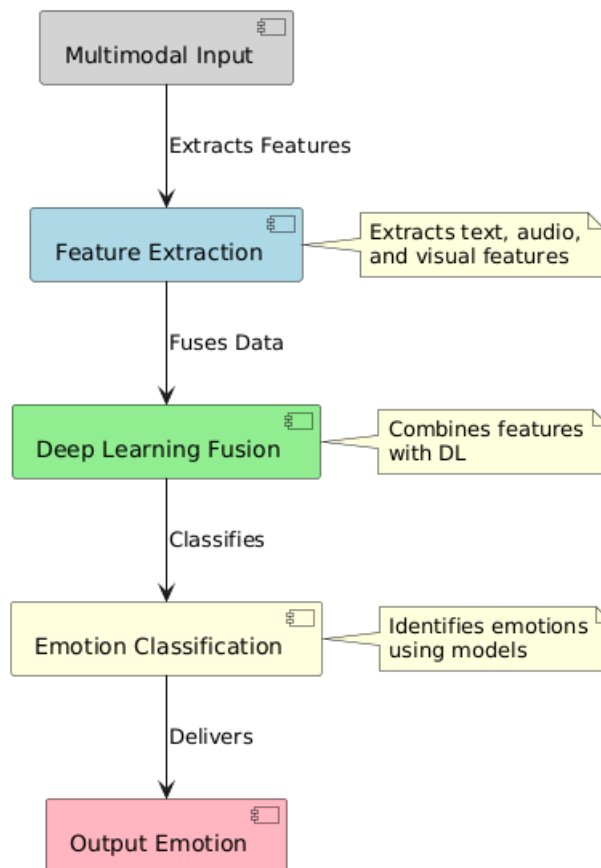


Figure 2.9: Multimodal Emotion Recognition with DL architecture

2.9.3 Advantages

1. Cross-modal signal integration.
2. Modular for flexibility.
3. Temporal modeling captures dynamic evolution of emotions over time.
4. Attention-based fusion highlights the most informative features across modalities.

2.9.4 Disadvantages

1. Dataset limitations.
2. Not tested for platform-specific optimization.
3. High computational cost due to processing multiple modalities and temporal modeling.
4. Complex architecture may be challenging to implement or scale in resource-constrained systems.

2.10 Applications and Reliability of Automatic Facial Emotion Recognition in Video Analysis

The paper titled “Applications and Reliability of Automatic Facial Emotion Recognition in Video Analysis” examines techniques for detecting emotions from facial expressions in videos and evaluates their accuracy, reliability, and practical applications. Unlike multimodal approaches, this study focuses primarily on face-based emotion recognition, leveraging visual cues such as facial landmarks, micro-expressions, and head movements to infer emotional states[10].

A key focus of this work is reliability analysis, which assesses how consistently facial emotion recognition systems perform across different conditions. Factors such as lighting, occlusions (e.g., glasses or masks), facial orientation, and subject variability can significantly impact accuracy. By systematically evaluating these influences, the study provides insights into the robustness and limitations of current face-based emotion detection systems.

2.10.1 Methodology

The methodology of “Applications and Reliability of Automatic Facial Emotion Recognition in Video Analysis” focuses on evaluating the reliability and performance of face-based emotion recognition systems while incorporating automated annotation for scalability. The approach emphasizes analyzing how well facial emotion detection works across varied video conditions and how consistent the system is in producing accurate emotion labels.

The first step involves facial feature extraction, where facial landmarks, expressions, micro-expressions, and head movements are detected using deep learning-based models or computer vision techniques such as CNNs. These features capture critical spatial cues indicative of different emotions.

Next, the system performs reliability analysis to measure consistency and accuracy under diverse scenarios. Factors considered include lighting variations, occlusions (e.g., glasses or masks), facial orientation, and inter-subject differences. Metrics such as detection accuracy, precision, recall, and consistency scores are computed to assess the robustness of the system.

2.10.2 Model Architecture

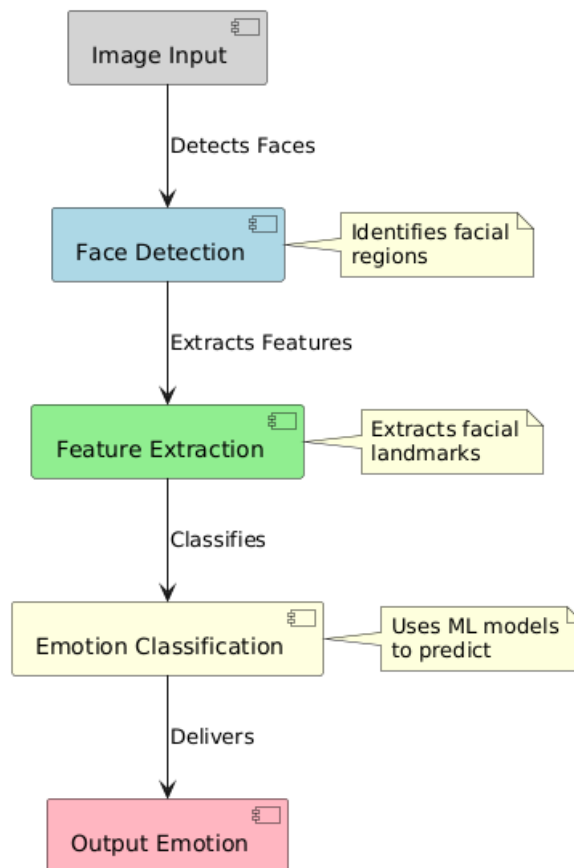


Figure 2.10: Facial Emotion Recognition architecture

2.10.3 Advantages

1. Reliable face-based detection.
2. Automated annotation.
3. Scalable for analyzing long video sequences efficiently.
4. Supports practical applications such as human-computer interaction, surveillance, education, and media analytics.

2.10.4 Disadvantages

1. Weak on occluded faces.
2. No cross-cultural study.
3. Sensitive to extreme lighting variations and head orientations.
4. Relies solely on visual cues, ignoring audio or textual emotional signals.

2.11 Emotion Recognition using multi-modal data and self-supervised learning

The paper titled “Emotion Recognition using Multi-Modal Data and Self-Supervised Learning” presents a modern approach to video-based emotion recognition by integrating multimodal data (video, audio, and text) with self-supervised learning techniques. Traditional supervised methods often require large amounts of labeled data, which can be expensive and time-consuming to collect. This study demonstrates how self-supervised learning can leverage unlabeled video data to improve the scalability and performance of emotion recognition systems[11].

The approach begins with feature extraction from multiple modalities. Visual features capture facial expressions, gestures, and body movements, audio features analyze pitch, tone, and speech patterns, and textual features extract semantic and sentiment information from transcripts or subtitles. By combining these modalities, the system achieves a holistic understanding of emotional states, compensating when one modality provides weak or ambiguous signals.

2.11.1 Methodology

The methodology of “Emotion Recognition using Multi-Modal Data and Self-Supervised Learning” integrates multimodal feature extraction with self-supervised learning techniques to achieve scalable and robust emotion recognition in real-world videos. The system begins by extracting visual, audio, and textual features from videos. Visual features, including facial expressions, micro-expressions, and body gestures, are captured using Convolutional Neural Networks (CNNs). Audio features such as pitch, tone, intensity, and speech dynamics are analyzed with Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks. Textual information from transcripts or subtitles is processed using Transformer-based natural language processing models to extract semantic and sentiment cues.

The key innovation lies in self-supervised learning, which allows the system to learn meaningful representations from unlabeled video data through auxiliary tasks, such as temporal order prediction, input feature reconstruction, or cross-modal alignment. These learned embeddings provide a strong foundation that can later be fine-tuned using smaller labeled datasets, significantly reducing the need for large-scale manual annotation.

2.11.2 Model Architecture

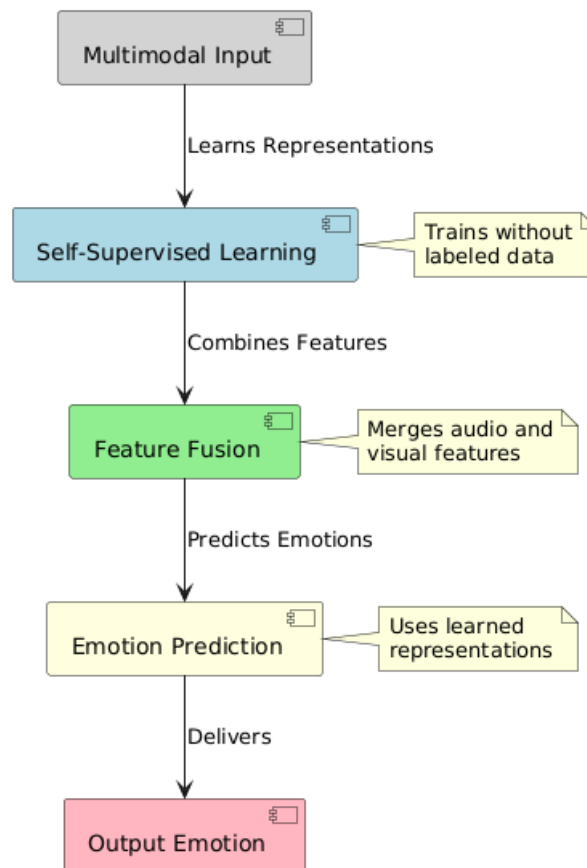


Figure 2.11: Multimodal Self-Supervised Emotion Recognition architecture

2.11.3 Advantages

1. Improved scaling for real videos.
2. Quality benchmarks.
3. Combines multimodal inputs (video, audio, text) for robust emotion recognition.
4. Reduces dependency on extensive labeled datasets through self-supervised learning.

2.11.4 Disadvantages

1. Large data demand.
2. Not optimized for live feeds.
3. High computational cost due to processing multiple modalities and training self-supervised models.
4. Complex architecture may be challenging to implement or scale in resource-constrained environments.

2.12 Real Time Emotion Recognition Using Image Classification

The paper titled “Real Time Emotion Recognition Using Image Classification” focuses on developing systems capable of detecting emotions instantaneously from facial images, enabling real-time applications. Unlike traditional offline or batch-processing emotion recognition systems, this study emphasizes low-latency detection, which is critical for interactive applications such as human-computer interaction, surveillance, and adaptive learning platforms[12].

The approach utilizes image classification techniques to analyze facial features and expressions in each frame of a video stream. Convolutional Neural Networks (CNNs) are typically employed to extract relevant spatial features from facial images, including micro-expressions, facial landmarks, and subtle muscle movements. These features are then fed into classification layers to predict the emotional state of the subject, providing a fast and efficient response suitable for live applications.

2.12.1 Methodology

The methodology of “Real Time Emotion Recognition Using Image Classification” focuses on detecting emotions instantaneously from facial images and integrating the results into real-time dashboards. The system captures video frames from a camera or live feed and applies image classification techniques to each frame. Convolutional Neural Networks (CNNs) are used to extract facial features, such as landmarks, micro-expressions, and subtle changes in facial muscles, which serve as the basis for predicting emotional states.

Once features are extracted, the system employs classification layers to label each frame with the corresponding emotion. The model is optimized for low-latency processing, ensuring that predictions are generated almost immediately, making the system suitable for interactive and real-time applications.

Additionally, the methodology incorporates dashboard integration, which visualizes detected emotions dynamically. This allows users to monitor trends, track emotional responses over time, and interpret the system’s outputs easily. Optimization techniques, such as lightweight CNN architectures and efficient inference pipelines, are applied to maintain high accuracy while ensuring fast response times. Overall, this methodology demonstrates a practical approach for real-time emotion recognition, combining speed, accuracy, and actionable visualization.

2.12.2 Model Architecture

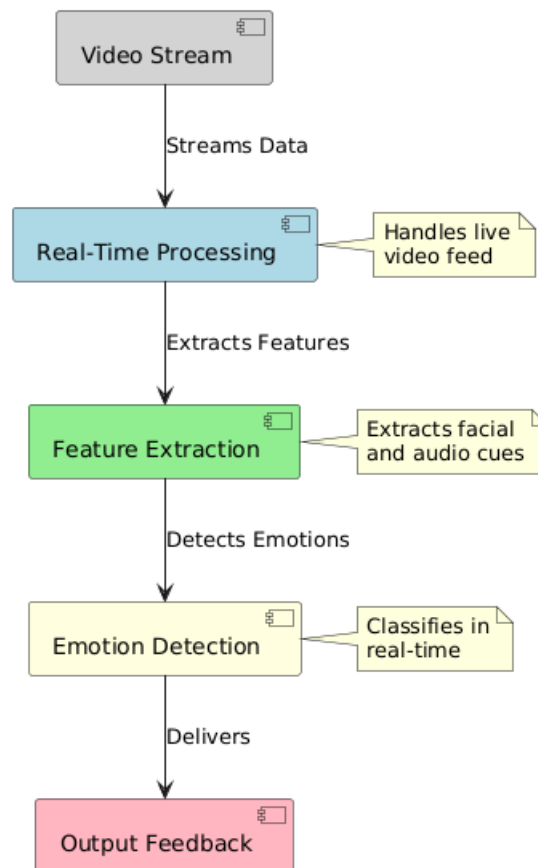


Figure 2.12: Real-Time Emotion Recognition architecture

2.12.3 Advantages

1. Real-time detection.
2. Dashboard integration.
3. Fast responses.
4. Fast response suitable for interactive applications.
5. Optimized for low-latency inference in live video streams.

2.12.4 Disadvantages

1. Focus on static images.
2. Needs extension for sequence emotion.
3. Performance may degrade on occluded or low-quality faces.
4. Limited consideration of multimodal cues like audio or text.

Chapter 3

PROPOSED SYSTEM

The proposed ZipClip system is designed as a multi-modal AI-driven framework for automated short-form content creation. The system begins with video ingestion, where raw long-form videos are uploaded or streamed into the platform. These videos are then processed by AI analysis modules, which detect key events, evaluate scene importance, and assign scores based on engagement, visual appeal, and narrative significance.

A central component of the system is highlight extraction, where the AI selects the most interesting or impactful moments from the video, ensuring that the resulting clips are engaging and coherent. The system also supports story reel creation from photos, enabling users to generate short video sequences or highlights from static images, combining motion, transitions, and effects.

To maximize utility across different platforms, ZipClip includes platform optimization, tailoring output formats, aspect ratios, and lengths according to the requirements of social media or streaming services. Additionally, the system features an interactive dashboard that allows users to preview clips, adjust parameters, track scoring metrics, and monitor content performance.

Overall, the proposed ZipClip system offers a fully automated, AI-powered pipeline that simplifies the process of creating short-form content from long videos or photo sequences. By combining multi-modal analysis, highlight extraction, platform-specific optimization, and user-friendly interactivity, ZipClip aims to streamline content creation while maximizing engagement and quality.

3.1 Process Overview

1. Project Planning

- a. Define Scope: Automate video to short clips with AI.
- b. Target Audience: Creators, educators, marketers.

- c. Timeline: Set milestones for development.
- d. Team: AI, web dev experts.

Requirements Gathering and Analysis

- a. User Needs: Efficient editing, customization.
- b. Analyze Feedback: Prioritize features.
- c. Requirements: Functional (AI analysis), non-functional (performance).

Design and Development.

- a. UI/UX: Intuitive dashboard.
- b. AI Integration: Modules for detection, scoring.
- c. Data: Video processing.
- d. Tech: React.js front, Node.js back, Python AI.

Testing and Quality Assurance

- a. Unit: Test modules.
- b. Integration: Ensure seamless flow.
- c. UAT: User testing.
- d. Performance: Load testing.

Deployment and Maintenance

- a. Deploy: Web app.
- b. Monitor: Feedback, support.
- c. Updates: Security, features.

3.1.1 Process Flow Diagram

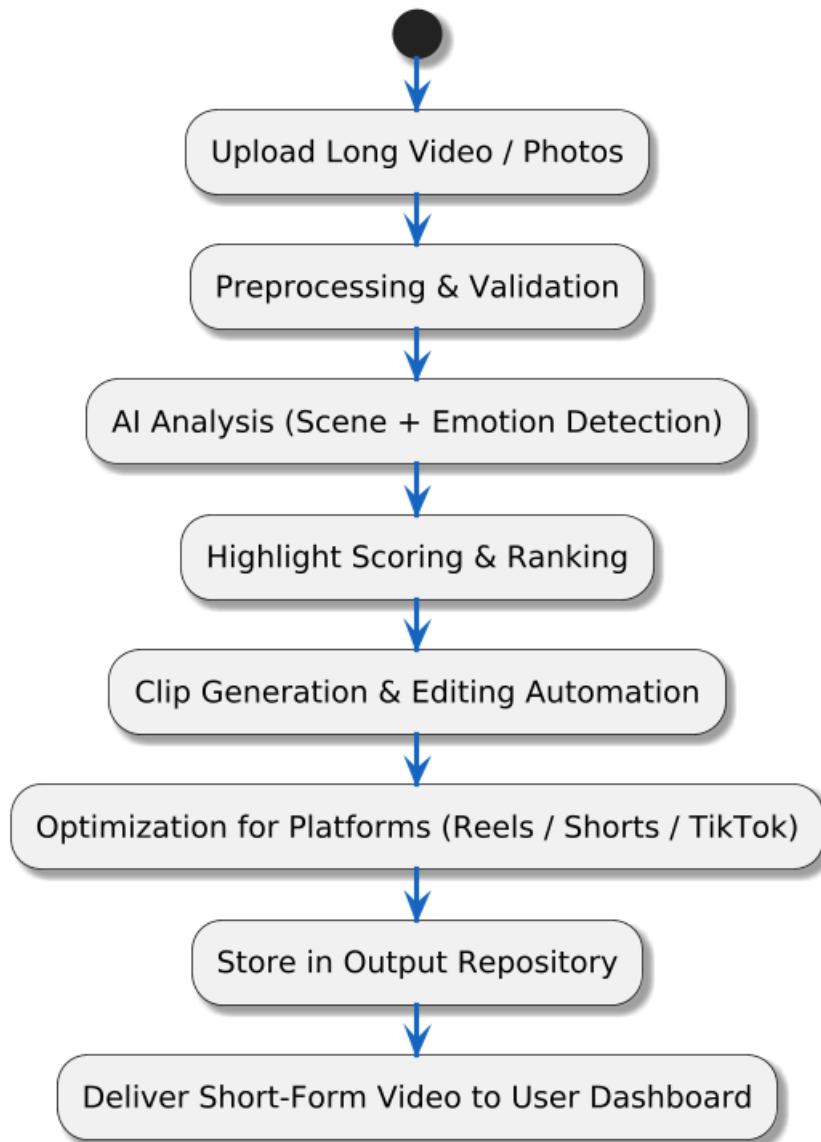


Figure 3.1: Process Flow Diagram

3.1.2 Architecture Diagram

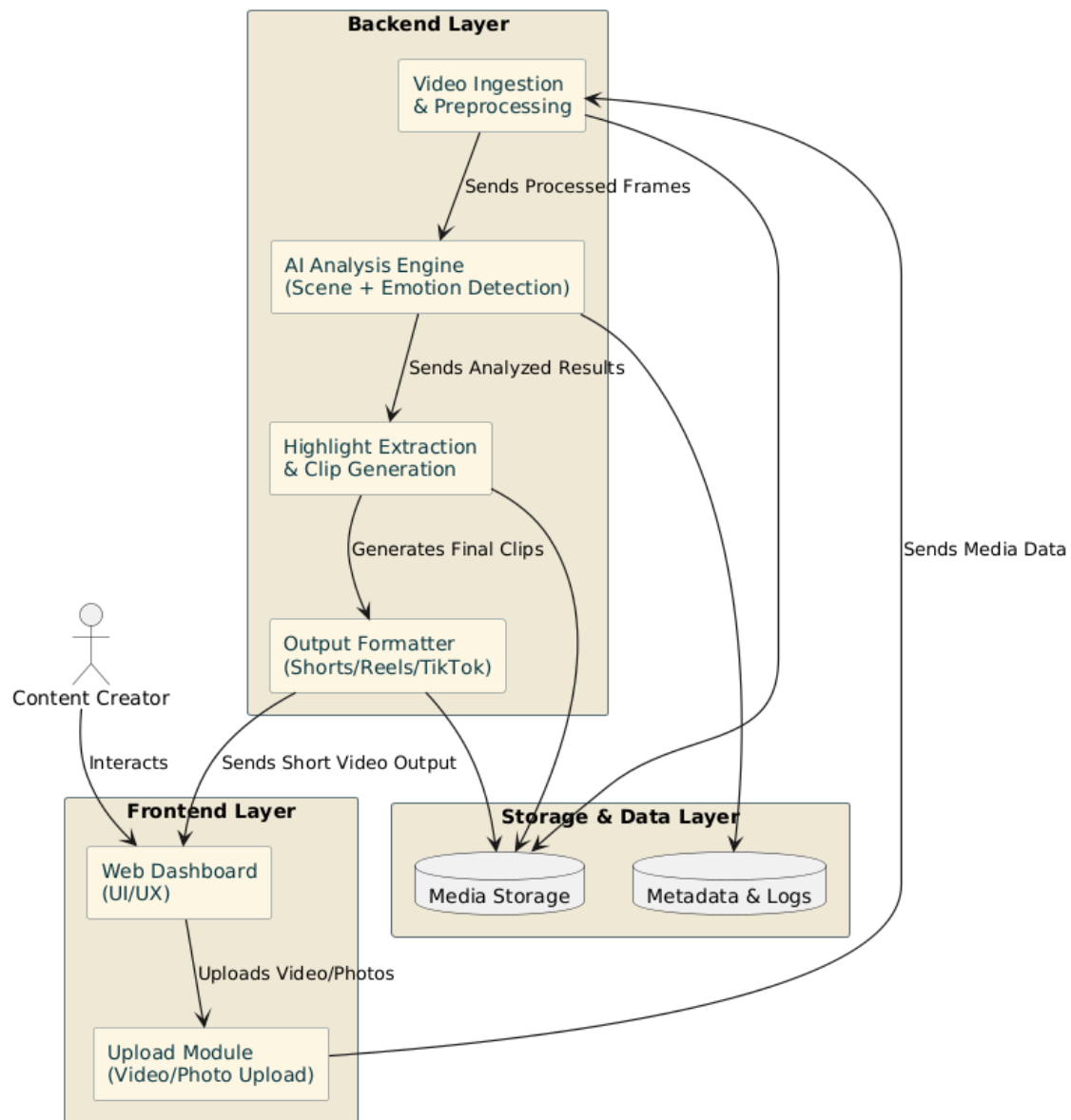


Figure 3.2: Architecture Diagram

3.1.3 Use case Diagram

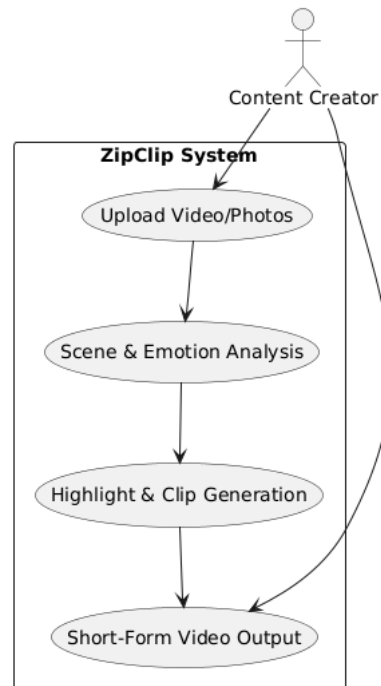


Figure 3.3: Use case diagram

3.1.4 Data flow diagram

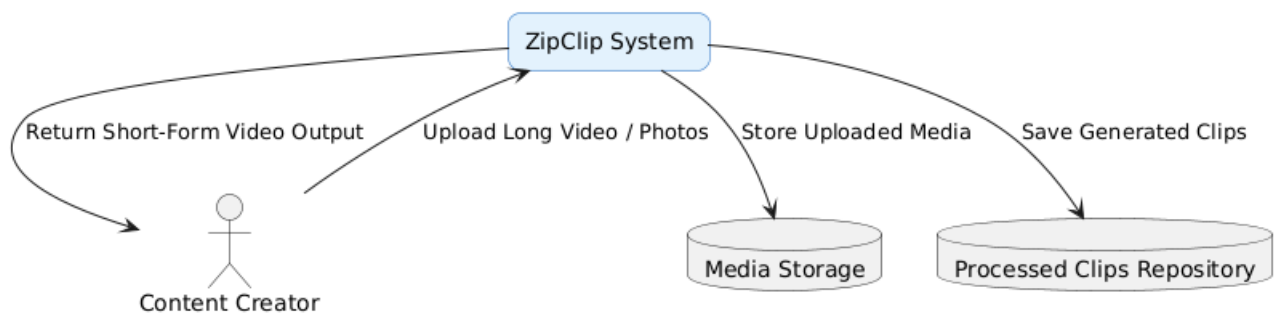


Figure 3.4: Data Flow Diagram Level 0

3.1.5 Data flow diagram

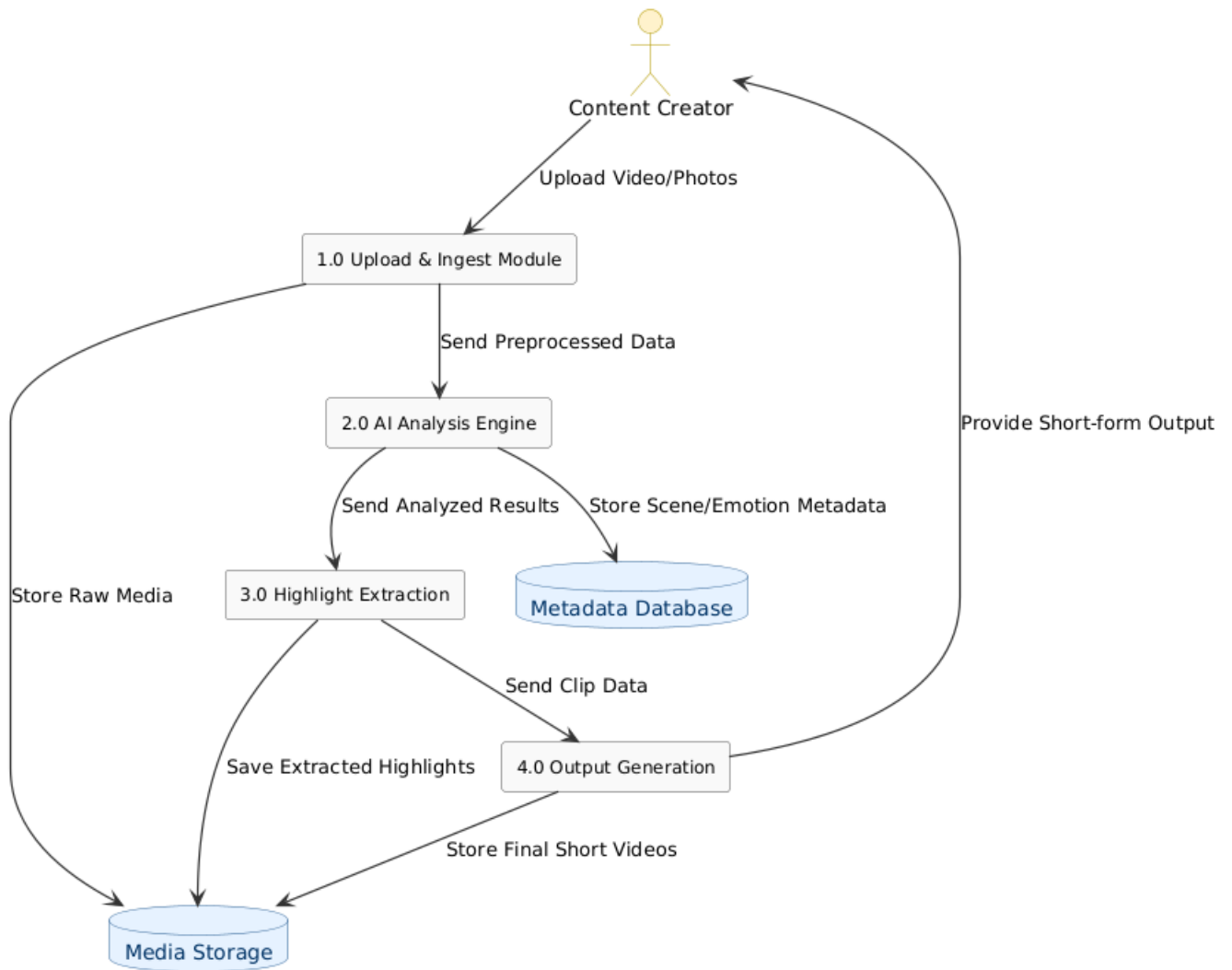


Figure 3.5: Data Flow Diagram Level 1-User

3.1.6 Class Diagram

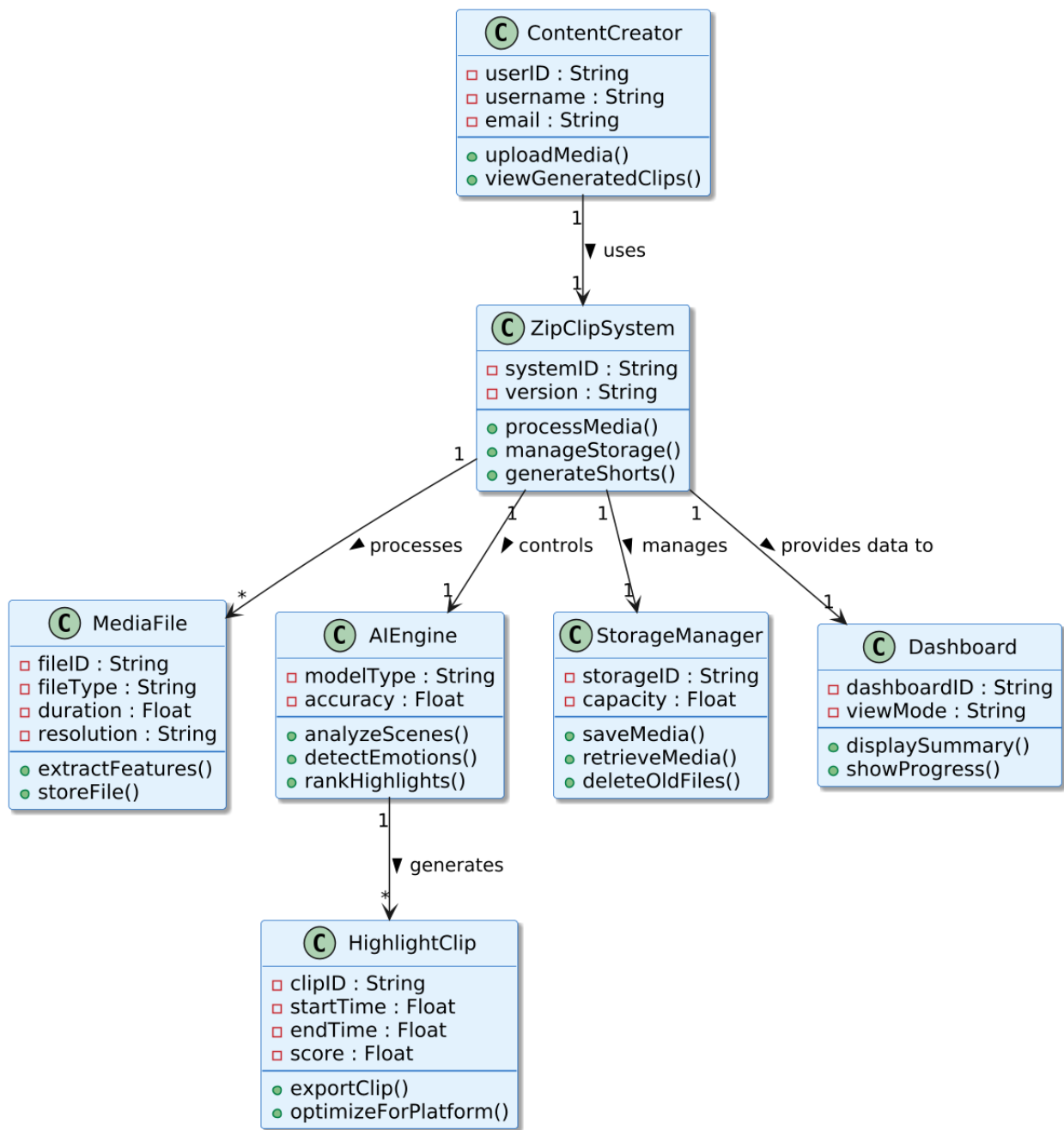


Figure 3.6: Class diagram

Chapter 4

CONCLUSION

The methodology of the ZipClip system revolves around automated, AI-driven short-form content creation. The process begins with video ingestion, where long-form videos are uploaded to the platform. The system then performs AI-based scene detection, identifying significant moments and segmenting the video into meaningful clips. In parallel, emotion analysis evaluates the affective content of each scene to prioritize engaging or impactful moments. These analyses are combined to enable smart clip generation, producing concise, coherent, and high-quality highlights suitable for short-form platforms.

To enhance usability, ZipClip incorporates story reel creation from photos, allowing static images to be transformed into dynamic video sequences with transitions and effects. The system also applies platform optimization, tailoring output formats, aspect ratios, and lengths to meet the requirements of social media platforms like YouTube Shorts, Instagram Reels, and TikTok. A modular and scalable architecture ensures that each component, from video analysis to highlight generation, can be independently updated or improved, supporting future enhancements such as auto-music selection, automatic subtitle generation, and multilingual support.

Finally, an interactive dashboard allows users to preview results, adjust parameters, and monitor clip scoring and performance metrics. This methodology emphasizes speed, scalability, and automation, reducing the need for complex and costly manual editing, while enabling creators, educators, and marketers to generate structured, engaging, and platform-ready content efficiently.

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